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Abstract—

*Index Terms—*ISAC, MTT, AAI

I. INTRODUCTION

Unmanned aerial vehicles (UAVs) have emerged as promising facilities in a variety of fields due to their flexibility, quick deployment, and high mobility [1]. By leveraging these attributes, UAVs outperform traditional stationary infrastructure by dynamically adjusting its position to maintain Line-of-Sight (LoS) connectivity and minimize signal blockage caused by buildings in the urban area. This makes UAVs particularly suited for real time detection and tracking of uncooperative targets such as hit-and-run vehicles, especially in Multi-Target Tracking (MTT) scenarios where multiple targets disperse across the area simultaneously [2], [3]. However, UAVs are constrained by limited onboard energy and computational capability. These constraints restrict their flight endurance, necessitate offloading of complex decision making to the base station (BS), and make an energy efficient trajectory design essential for prolonged tracking mission [4]. Moreover, conventional approaches that employ separate hardware and dedicated spectrum for sensing and communication further exacerbate the energy burden []. To overcome these limitations, integrated sensing and communication (ISAC) unifies radar sensing and communication functionalities into a single transceiver, thereby improving spectral efficiency, reducing hardware complexity, and freeing up energy that would otherwise be consumed by operating separate sensing and communication modules [5]. We propose a multi-UAV enabled ISAC system for MTT.

II. SYSTEM MODEL

A. Architecture

We consider a multi-UAV assisted multi-target tracking system coordinated by a BS located at the origin (0,0). The system comprises a set $\mathcal{U} = \{1, 2, \dots, U\}$ of UAVs $u \in \mathcal{U}$ and a set $\mathcal{K} = \{1, 2, \dots, K\}$ of mobile ground targets $k \in \mathcal{K}$ in simulated environment. The operational area additionally contains a set $\mathcal{Z} = \{1, 2, \dots, Z\}$ of predefined critical zones, representing sensitive civilian areas such as schools or hospitals that demand heightened surveillance. The BS operates an AAI framework, which operates a perception-reasoning-action loop to continuously monitor evolving mission conditions such as UAV energy levels, communication link quality, and target proximity to critical zones, and dynamically reassign targets to UAVs. The operational time horizon T is divided into N equal time slots, denoted by $\mathcal{T} = \{1, 2, \dots, T\}$. Each time slot is indexed by $t \in \mathcal{T}$, and each time slot has constant duration of $\delta = T/N$ [1], [6]–[8]. Each time slot is further divided into two sub slots, where the UAV

performs target sensing during the first sub slot and transmits the acquired data to the BS during the second sub slot [6]–[8]. Therefore, the time allocation within each time slot must satisfy $\tau_s + \tau_c = \delta$. Each UAV flies at a fixed altitude H and a maximum speed v_{max} and is equipped with [list external sensing device](#) []. To maximize sensing Signal-to-Noise (SNR), each UAV is assigned to track at most one target per time slot, concentrating UAV's full antenna gain toward a single target and avoiding transmit power split across multiple direction. The 2-D Cartesian coordinate of u -th UAV at time slot t is denoted as $\mathbf{q}_{u,t} = [x_{u,t}, y_{u,t}]^\top$. The state of k -th target at time slot t is defined as $\mathbf{S}_{k,t} = [x_{k,t}, y_{k,t}, \dot{x}_{k,t}, \dot{y}_{k,t}]^\top$, which comprises its 2-D Cartesian position and velocity component [4], [9]. For distance calculations, we explicitly denote the 2-D position of the k -th target as $\mathbf{q}_{k,t} = [x_{k,t}, y_{k,t}]^\top$, which corresponds to the first two elements of $\mathbf{S}_{k,t}$.

Before UAV tracking begins, the BS collects the initial dispatch position of each UAV u and target k . The per-time-slot operational procedure is as follows:

- 1) At the beginning of each time slot, the u -th UAV emits a radar signal for target detection.
- 2) After receiving radar echoes, the UAV locally extracts the raw measurement vector $z_{u,k,t}$ and computes the local posterior state estimate $S_{u,k,t|t}$ and its corresponding covariance matrix $P_{u,k,t|t}$ using the Extended Kalman Filter (EKF).
- 3) Following detection, the UAV transmits its current position $\mathbf{q}_{u,t}$, remaining energy level $E_{u,t}$, and locally processed target estimates to the BS. If the target was not detected within the u -th UAV's sensing range, the UAV instead transmits a binary absence signal $\alpha_{u,k,t} = 0$ to minimize communication overhead.
- 4) Upon receiving the local target state estimates from all active UAVs, the BS fuses them into a global state estimate $\tilde{\mathbf{S}}_{k,t}$ and covariance $\tilde{\mathbf{P}}_{k,t}$ for each target k , as detailed in Section II-G.
- 5) Based on the global perceived state Φ_t , the AAI framework computes the dynamic parameters, including the tracking accuracy threshold $\epsilon_{k,t}$ and the priority weight $W_{k,t}$ for each target.
- 6) The BS broadcasts the parameters and the fused target state back to the active UAVs.
- 7) Upon receiving the transmission, each UAV determines its next waypoint $\mathbf{q}_{u,t+1}$ and moves to its computed assigned waypoint.

B. State Space Model

The state space representation of the target is described as [9], [10]:

$$\mathbf{S}_{k,t+1} = \mathbf{F}\mathbf{S}_{k,t} + \mathbf{w}_{k,t}, \quad (1)$$

where $\mathbf{F} \in \mathbb{R}^{4 \times 4}$ is the transition matrix defined as

$$\mathbf{F} = \begin{bmatrix} 1 & 0 & \delta & 0 \\ 0 & 1 & 0 & \delta \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (2)$$

and $\mathbf{w}_{k,t}$ is the corresponding zero-mean Gaussian random process noise with covariance matrix

$$\mathbf{Q} = \begin{bmatrix} \delta^4/4 & 0 & \delta^3/2 & 0 \\ 0 & \delta^4/4 & 0 & \delta^3/2 \\ \delta^3/2 & 0 & \delta^2 & 0 \\ 0 & \delta^3/2 & 0 & \delta^2 \end{bmatrix} \sigma_w^2, \quad (3)$$

where σ_w^2 is the noise variance.

C. Measurement Model

We introduce a binary variable $\alpha_{u,k,t}$ that indicate whether UAV u senses target k in time slot t , $\alpha_{u,k,t} = 1$, otherwise $\alpha_{u,k,t} = 0$.

Let us denote the measurement vector by $\mathbf{z}_{u,k,t}$ as follows [9], [10]:

$$\mathbf{z}_{u,k,t} = \alpha_{u,k,t} \cdot (h(\mathbf{S}_{k,t}) + \mathbf{v}_{u,k,t}), \quad (4)$$

where $h(\mathbf{S}_{k,t}) = [\sqrt{(x_{k,t} - x_{u,t})^2 + (y_{k,t} - y_{u,t})^2}, \arctan \frac{y_{k,t} - y_{u,t}}{x_{k,t} - x_{u,t}}]$ denotes the measurement function that maps $\mathbf{S}_{k,t}$ to $\mathbf{z}_{u,k,t}$ and $\mathbf{v}_{u,k,t} = [\mathbf{v}_{u,k,t}^r, \mathbf{v}_{u,k,t}^\theta]^\top$ denotes the measurement noise vector, which consists of the range measurement noise $\mathbf{v}_{u,k,t}^r$ and the angle measurement noise $\mathbf{v}_{u,k,t}^\theta$ at time slot t . $\mathbf{v}_{u,k,t}^r$ and $\mathbf{v}_{u,k,t}^\theta$ are zero-mean Gaussian noise components with variances $(\sigma_{u,k,t}^r)^2$ and $(\sigma_{u,k,t}^\theta)^2$, respectively.

The relationship between the variance of measurement noise and the signal-to-noise ratio (SNR) can be determined as [9], [10]:

$$(\sigma_{u,k,t}^\bullet)^2 = \frac{(\sigma_0^\bullet)^2}{\gamma_{u,k,t}^{\text{sen}}}, \quad (5)$$

where $\bullet \in \{r, \theta\}$. $(\sigma_0^\bullet)^2$ denotes the reference value of the corresponding measurement noise variance. The SNR of the radar link can be formulated as [8]–[10]:

$$\gamma_{u,k,t}^{\text{sen}} = \frac{p_{u,t} G_t G_r \lambda^2 \sigma \tau_s}{N_0 (4\pi)^3 d_{u,k,t}^4}, \quad (6)$$

where $p_{u,t}$ is the transmit power of UAV, G_t and G_r are the antenna gains of transmitter and receiver, respectively, $\lambda = \frac{c}{f_c}$ is wavelength, c is the speed of light and f_c is the carrier frequency, σ denotes target radar cross-section (RCS), τ_s is the dwell time for tracking the target, N_0 is the power of additive white Gaussian noise and $d_{u,k,t} = \sqrt{\|\mathbf{q}_{u,t} - \mathbf{q}_{k,t}\|^2 + H^2}$ is the distance between UAV u and target k .

sensing energy [6]

$$E_{u,t}^{\text{sen}} = p_{u,t} \tau_s. \quad (7)$$

D. Extended Kalman Filter

Due to the nonlinear relationship between the target state to be estimated and the measurement, the EKF is adopted [1], [2], [4], [9], [10]. The predicted state $\hat{\mathbf{S}}_{k,t|t-1}$ and covariance matrix $\hat{\mathbf{P}}_{k,t|t-1}$ of the target can be given as [1], [9], [10]:

$$\hat{\mathbf{S}}_{k,t|t-1} = \mathbf{F} \mathbf{S}_{k,t-1|t-1}, \quad (8)$$

$$\hat{\mathbf{P}}_{k,t|t-1} = \mathbf{F} \mathbf{P}_{k,t-1|t-1} \mathbf{F}^\top + \mathbf{Q}, \quad (9)$$

where $\mathbf{S}_{k,t|t}$ and $\mathbf{P}_{k,t|t}$ are the posterior estimate and the corresponding covariance matrix.

the optimal Kalman gain $\mathbf{K}_{u,k,t}$ is defined as [1], [9], [10]:

$$\mathbf{K}_{u,k,t} = \hat{\mathbf{P}}_{k,t|t-1} \mathbf{H}_{u,k,t}^\top (\mathbf{H}_{u,k,t} \hat{\mathbf{P}}_{k,t|t-1} \mathbf{H}_{u,k,t}^\top + \mathbf{R}_{u,k,t})^{-1}, \quad (10)$$

where $\mathbf{H}_{u,k,t}$ is the observation matrix defined as the Jacobian matrix of the measurement function $h(\cdot)$:

$$\mathbf{H}_{u,k,t} = \begin{bmatrix} \frac{\Delta x}{r^2} & \frac{\Delta y}{r^2} & 0 & 0 \\ -\frac{\Delta y}{r^2} & \frac{\Delta x}{r^2} & 0 & 0 \end{bmatrix}, \quad (11)$$

where $\Delta x = x_{k,t} - x_{u,t}$, $\Delta y = y_{k,t} - y_{u,t}$, and $r = \sqrt{\Delta x^2 + \Delta y^2}$. $\mathbf{R}_{u,k,t}$ is the covariance matrix of the measurement error defined as

$$\mathbf{R}_{u,k,t} = \begin{bmatrix} (\sigma_t^r)^2 & 0 \\ 0 & (\sigma_t^\theta)^2 \end{bmatrix}. \quad (12)$$

The posterior state estimate $\mathbf{S}_{u,k,t|t}$ and the posterior covariance matrix $\mathbf{P}_{u,k,t|t}$ are given as [1], [9], [10]:

$$\mathbf{S}_{u,k,t|t} = \hat{\mathbf{S}}_{k,t|t-1} + \mathbf{K}_{u,k,t} (\mathbf{z}_{u,k,t} - h(\hat{\mathbf{S}}_{k,t|t-1})), \quad (13)$$

$$\mathbf{P}_{u,k,t|t} = (\mathbf{I} - \mathbf{K}_{u,k,t} \mathbf{H}_{u,k,t}) \hat{\mathbf{P}}_{k,t|t-1}, \quad (14)$$

where $\mathbf{I} \in \mathbb{R}^{4 \times 4}$ denotes the unity matrix.

E. Communication Model

The probability of an LoS connection is given by [11]:

$$P_{u,t}^{\text{LoS}} = (1 + c_1 \exp[-c_2 (\theta_{u,t} - c_1)])^{-1}, \quad (15)$$

where $\theta_{u,t} = \arctan(\frac{H}{d_{u,t}})$ denotes the elevation angle between UAV u and BS, $d_{u,t} = \|\mathbf{q}_{u,t} - \mathbf{q}_b\|$ denotes the distance between UAV u and BS at time slot t , and c_1 and c_2 are environment-dependent constants. The probability of having an NLoS connection is given by $P_{u,t}^{\text{NLoS}} = 1 - P_{u,t}^{\text{LoS}}$.

The average channel power gain between UAV u and BS at time slot t can be expressed as [11]:

$$h_{u,t}^c = \frac{(P_{u,t}^{\text{LoS}} \eta^{\text{LoS}} + P_{u,t}^{\text{NLoS}} \eta^{\text{NLoS}}) \lambda^2}{(4\pi)^2 d_{u,t}^2}, \quad (16)$$

where η^{LoS} and η^{NLoS} denote the additional average loss terms that capture environment-dependent attenuation for LoS and NLoS links, respectively.

The communication rate of UAV u to BS at time slot t can be expressed as [9]:

$$R_{u,t}^c = \alpha_{u,k,t} \tau_c B_b \log_2(1 + \gamma_{u,t}^c), \quad (17)$$

where $\tau_c = \delta - \tau_s$ is the communication time and $\gamma_{u,t}^c = \frac{p_{u,t} h_{u,t}^c}{N_0}$ is the SNR of the communication link. energy [6]

$$E_{u,t}^{\text{comm}} = p_{u,t} \tau_c. \quad (18)$$

F. Energy Model

The propulsion power consumption of the UAV is given by, $P^{\text{move}} = P_0 \left(1 + \frac{3\|v\|^2}{u_{\text{tip}}^2}\right) + P_1 \left(\sqrt{1 + \frac{\|v\|^4}{4v_0^4}} - \frac{\|v\|^2}{2v_0^2}\right)^{\frac{1}{2}} + \frac{1}{2}d_0\rho s_0\nu\|v\|^3$, where P_0 and P_1 represent the blade profile and induced powers in the hovering state, respectively. u_{tip} denotes the rotor blade tip speed of the UAV, v_0 is the mean rotor induced velocity in the hovering state, d_0 denotes the fuselage drag ratio, ρ is the density of air, s_0 represents the rotor solidity, and ν denotes the area of the rotor disk.

The energy consumption of the UAV during its flight from $\mathbf{q}_{u,t-1}$ to $\mathbf{q}_{u,t}$ is

$$E_{u,t}^{\text{move}} = P^{\text{move}}\delta. \quad (19)$$

The total energy of UAV u at time slot t can be expressed as

$$E_{u,t}^{\text{tot}} = E_{u,t}^{\text{sen}} + E_{u,t}^{\text{comm}} + E_{u,t}^{\text{move}}. \quad (20)$$

G. Fusion

The BS combines all the local estimates as [1]:

$$\tilde{\mathbf{S}}_{k,t} = \tilde{\mathbf{P}}_{k,t} \left(\sum_{u \in \mathcal{U}} \omega_u (\mathbf{P}_{u,k,t|t})^{-1} \mathbf{S}_{u,k,t|t} \right), \quad (21)$$

$$\tilde{\mathbf{P}}_{k,t} = \left(\sum_{u \in \mathcal{U}} \omega_u (\mathbf{P}_{u,k,t|t})^{-1} \right)^{-1}, \quad (22)$$

where ω_u denotes the weighting coefficients and $\sum_{u \in \mathcal{U}} \omega_u = 1$.

$$[\{\epsilon_{k,t}\}_{k \in \mathcal{K}}, \{W_{k,t}\}_{k \in \mathcal{K}}, \{p_{u,t}\}_{u \in \mathcal{U}}] = f_{\text{AAI}}(\Phi_t), \quad (23)$$

where $\epsilon_{k,t}$ denotes the tracking accuracy threshold, $W_{k,t}$ denotes the priority weight for target k . Φ_t represents the global perceived state by the AAI framework at time slot t , defined as:

$$\Phi_t \triangleq \left(\{\tilde{\mathbf{S}}_{k,t}, \tilde{\mathbf{P}}_{k,t}, d_{k,t}^{\mathcal{Z}}\}_{k \in \mathcal{K}}, \{E_{u,t}\}_{u \in \mathcal{U}} \right), \quad (24)$$

where $d_{k,t}^{\mathcal{Z}}$ denotes the minimum distance between the target k and the nearest critical zone.

H. Tracking Accuracy Metric

Assume that $\mathbf{S}_{k,t|t}(\mathbf{Z}_{k,t})$ is the target state estimate based on the measurement set $\mathbf{Z}_{k,t}$ with $\mathbf{Z}_{k,t} = \{\mathbf{z}_{u,k,t} | \forall u \in \mathcal{U}\}$, we will have the following Cramér-Rao inequality [1], [12]:

$$\begin{aligned} \mathbb{E}_{\mathbf{S}_{k,t}, \mathbf{Z}_{k,t}} \left\{ [\mathbf{S}_{k,t|t}(\mathbf{Z}_{k,t}) - \mathbf{S}_{k,t}] [\mathbf{S}_{k,t|t}(\mathbf{Z}_{k,t}) - \mathbf{S}_{k,t}]^{\top} \right\} \\ \geq \mathbf{J}^{-1}(\mathbf{S}_{k,t}) = \mathbf{C}(\mathbf{S}_{k,t}), \end{aligned} \quad (25)$$

where $\mathbb{E}_{\mathbf{S}_{k,t}, \mathbf{Z}_{k,t}} \{\cdot\}$ denotes the expectation operation with respect to the target state and the measurements, $\mathbf{J}(\mathbf{S}_{k,t})$ is the Bayesian Fisher information matrix (BFIM) about the target state, and the posterior Cramér-Rao lower bound (PCRLB) matrix $\mathbf{C}(\mathbf{S}_{k,t})$ is the inverse of the BFIM.

As for the BFIM $\mathbf{J}(\mathbf{S}_{k,t})$, we have the following expression [1], [12]:

$$\begin{aligned} \mathbf{J}(\mathbf{S}_{k,t}) &= [\mathbf{F}^{-1}]^{\top} \mathbf{J}(\mathbf{S}_{k,t-1}) \mathbf{F}^{-1} + \mathbf{Q}^{-1} \\ &+ \sum_{u=1}^U \alpha_{u,k,t} (\mathbf{H}_{u,k,t})^{\top} (\mathbf{R}_{u,k,t})^{-1} \mathbf{H}_{u,k,t}. \end{aligned} \quad (26)$$

The performance metric function that focusing on target tracking accuracy is defined as [1]:

$$\mathcal{F}_{k,t} = \text{tr}(\mathbf{\Lambda} \cdot \mathbf{J}^{-1}(\mathbf{S}_{k,t}) \cdot \mathbf{\Lambda}^{\top}), \quad (27)$$

where $\mathbf{\Lambda} = \text{diag}(1, 1, \delta, \delta)$ denotes the normalization matrix.

III. PROBLEM FORMULATION

The optimization problem is formulated as follows:

$$\mathcal{P} : \min_{\mathbf{\Omega}} \sum_{t=1}^T \sum_{u=1}^U E_{u,t}^{\text{tot}} \quad (28a)$$

$$\text{s.t. } x_{u,t}, x_{k,t} \in [x_{\min}, x_{\max}] \quad (28b)$$

$$y_{u,t}, y_{k,t} \in [y_{\min}, y_{\max}] \quad (28c)$$

$$\|\mathbf{q}_{u,t} - \mathbf{q}_{u,t-1}\| \leq v_{\max} \delta \quad (28d)$$

$$\|\mathbf{q}_{i,t} - \mathbf{q}_{j,t}\| \geq d_{\min}, \quad \forall i, j \in \mathcal{U}, i \neq j \quad (28e)$$

$$\sum_{u \in \mathcal{U}} \alpha_{u,k,t} \geq 1 \quad (28f)$$

$$\gamma_{u,k,t}^{\text{sen}} \geq \gamma_{\text{th}}^{\text{sen}} \quad (28g)$$

$$R_{u,t}^c \leq R_{\text{th}}^c \quad (28h)$$

$$\mathcal{F}_{k,t} \leq \epsilon_{k,t} \quad (28i)$$

where $\mathbf{\Omega}$ denotes the set of UAV trajectories to be optimized.

IV. ALGORITHM

A. AAI

The BS utilizes an AAI framework to drive the perception-reasoning-action loop. The AAI operates as a high-level policy modulator that dynamically evaluates the operational environment to shape the tracking policies of the active UAVs.

1) *Perception Stage*: The BS continuously aggregate the environment information. It extract the UAV's resource constraints, communication link qualities, and the evolving spatial relationship between mobile targets and sensitive areas. This structured perception enables the AAI to capture the current operational context, and provide a foundation for subsequent trajectory planning decisions.

2) *Reasoning Stage*: Based on the perceived global telemetry, the AAI computes three dynamic parameters to modulate UAVs' response. The tracking accuracy threshold $\epsilon_{k,t}$ specifies an upper bound on the tracking error and is tightened when targets approaches critical zones. The target priority weight $W_{k,t}$ encodes the relative importance of each target. The ISAC transmit power $p_{u,t}$ is allocated per UAV based on balance sensing quality against energy consumption, with higher power assigned when channel conditions or target distances demand stronger echoes.

3) *Action Stage*: The AAI broadcast the computed parameters $\epsilon_{k,t}, W_{k,t}, p_{u,t}$ to each UAVs. The subsequent waypoint $q_{u,t+1}$ are then resolved by the UAV agents adapting their local policies to maximize the modulated reward functions during environmental interaction.

B. Dec-POMDP

We formulate the multi-UAV target trajectory optimization problem as Decentralized Partially Observable Markov Decision Process (Dec-POMDP) defined as a tuple $\{\mathcal{U}, \mathcal{S}, \{\mathcal{O}_u\}, \{\mathcal{A}_u\}, \mathcal{P}, \mathcal{R}, \gamma\}$, which consists of a set of agents, global state space, local observation space, action space, state transition probability distribution, reward function, and discount factor [7], [13].

1) *Global State Space $\mathcal{S}$* : In a Dec-POMDP, the global state is fully observed by the centralized critic at each time slot t during the training phase. The global state is a combination of all UAVs' current position $\mathbf{q}_{u,t}$ and remaining energy $E_{u,t}$, as well as targets' predicted states $\tilde{\mathbf{S}}_{k,t}$, tracking accuracies $\mathcal{F}_{u,k,t}$, distance to the nearest critical zone $d_{k,t}^Z$, and the AAI's dynamic parameters $\epsilon_{k,t}$ and $W_{k,t}$. Let s_t be the global state at time slot t , then

$$s_t = \{\{\mathbf{q}_{u,t}, E_{u,t}\}_{u \in \mathcal{U}}, \{\tilde{\mathbf{S}}_{k,t}, \mathcal{F}_{k,t}, d_{k,t}^Z, \epsilon_{k,t}, W_{k,t}\}_{k \in \mathcal{K}}\}, \quad (29)$$

where the remaining energy of each UAV is bounded by its battery capacity, i.e., $E_{u,t} \in [0, E_{\max}] \forall u \in \mathcal{U}, \forall t$.

2) *Local Observation Space $\mathcal{O}_u$* : In a Dec-POMDP, each agent cannot fully observe the global environment. Each u -th UAV is viewed as an agent in the system, and its local observation is a combination of its own position $\mathbf{q}_{u,t}$, remaining energy $E_{u,t}$, the relative distance to its assigned target $\Delta \mathbf{q}_{u,k,t}$, the target's predicted state $\tilde{\mathbf{S}}_{k,t}$, and the specific AAI parameters broadcasted to it. Let $o_{u,t}$ be the local observation of the u -th UAV at time slot t , then

$$o_{u,t} = (\mathbf{q}_{u,t}, E_{u,t}, \Delta \mathbf{q}_{u,k,t}, \tilde{\mathbf{S}}_{k,t}, \epsilon_{k,t}, W_{k,t}). \quad (30)$$

3) *Action Space $\mathcal{A}_u$* : In a Dec-POMDP, each agent takes the next action based on its local observation and current state. The action that each UAV agent can take is a 2-D vector indicating its next waypoint. Due to mobility constraint imposed by maximum UAV speed, the action space is bounded. let $a_{u,t}$ be the action of the u -th UAV at time slot t , then

$$a_{u,t} = \Delta \mathbf{q}_{u,t} = [\Delta x_{u,t}, \Delta y_{u,t}]^\top, \forall u \in \mathcal{U}, \forall t, \quad (31)$$

where $\|\Delta \mathbf{q}_{u,t}\| \leq v_{\max} \delta$. The next waypoint is given by $\mathbf{q}_{u,t+1} = \mathbf{q}_{u,t} + \Delta \mathbf{q}_{u,t}$.

4) *State Transition Probability Distribution $\mathcal{P}$* : In a Dec-POMDP, the state transition follows a Markov chain, where the next state depends on current global state and the joint action taken by all agents. In this work, while the UAV positions updates deterministically as $\mathbf{q}_{u,t+1} = \mathbf{q}_{u,t} + \Delta \mathbf{q}_{u,t}$ and remaining energy depletes as $E_{u,t+1} = E_{u,t} - E_{u,t}^{\text{tot}}$. The overall state transition is stochastic due to target motion noise $\mathbf{w}_{k,t}$. Therefore, the transition probability needs to be learned implicitly by the MADRL agent through interaction with the environment.

5) *Reward Function $\mathcal{R}$* : In a Dec-POMDP, the reward function defines the objective that each agent optimizes through interaction with the environment. In this work, the reward is designed to minimize total energy consumption while penalizing tracking accuracy degradation and failure of target detection. The global reward r_t is defined as

$$r_t = - \sum_{u \in \mathcal{U}} E_{u,t}^{\text{tot}} - \sum_{k \in \mathcal{K}} W_{k,t} (\lambda_1 \mathcal{F}_{k,t} + \lambda_2 \prod_{u \in \mathcal{U}} (1 - \alpha_{u,k,t})) - \lambda_3 \mathcal{C}_t \quad (32)$$

where $\lambda_1, \lambda_2, \lambda_3$ are the weighting coefficients, $\mathcal{C}_t$ is the total collision count at time slot t . The reward function consists of three distinct penalty components. The first term acts as a baseline penalty for total energy consumption of the entire UAV swarm. The second term is the collective tracking and target loss penalty scaled by AAI assigned priority $W_{k,t}$. The third term penalize the total number of constraint violation in the swarm.

6) *Discount Factor γ* : In a Dec-POMDP, the discount factor $\gamma \in [0, 1]$ determines valuation for future rewards relative to immediate rewards for the agent. In this work, γ is set within the range $[0.9, 0.99]$ to emphasize long-term trajectory planning while maintaining stable training convergence.

C. MADRL

V. PERFORMANCE EVALUATION

Parameter	Value
Maximum UAV velocity, $v_{\max}$	10 m/s
UAV flight altitude, H	100 m
Dwell time, δ	1 s
Maneuverability noise variance, σ_w^2	5 (m/s ²) ²
Measurement noise variance, $(\sigma_0^w)^2$	10 m ²
Measurement noise variance, (σ_0^θ)	10 ⁻⁴ rad ²
Transmit power, p	15 W
Transmitter gains G_t , receiver gains, G_r	20 dBi, 30 dBi
SNR threshold, $\gamma_{\text{th}}^{\text{sen}}$	13 dB,
Data communication threshold, R_{th}^c	1.218 bps/Hz
Target RCS, σ	1 m ²
Gaussian white noise, N_0	-110 dBmW
Environment dependent constant, c_1, c_2	11.9, 0.13
Channel parameter, $\eta^{\text{LoS}}, \eta^{\text{NLoS}}$	1 dB, 20 dB
Blade profile power, P_0 , induced power, P_1	3.4 W, 20 W
Rotor blade tip speed, u_{tip}	60 m/s
Mean rotor induced velocity, v_0	5.4 m/s
Fuselage drag ratio, d_0	0.3
Density of air, ρ	1.225 kg/m ³
Rotor solidity, s_0	0.03
Area of the rotor disk, ν	0.28 m ²

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